

PENDING TOPIC OF MODULE-I

- 1. Human Population**
- 2. Sustainable development**
- 3. Sustainable development in INDIA**
- 4. Sustainable development in UN**
- 5. Concept of Circular Economy**
- 6. Concept of Entrepreneurship**

HUMAN POPULATION



□ INTRODUCTION

- Population - "all the inhabitants of a particular town, area, or country"
- The human population is the total number of living humans on Earth
- Environment means "the sum total of all surroundings of a living organism, including natural forces and other living things, which provide conditions for development and growth as well as of danger and damage"

▣ Human Population

▣ Environment



POPULATION DENSITY

Population Density

Population Density is a measure of how many individuals of a given species are found in a given area.

There are two types of density:

- 1) **Crude density** is the total number of individuals divided by the total area of the entire habitat.
- 2) **Ecological density** is the total number of individuals divided by the total useable area in the habitat.

The equation for population density (D) is:

$$D = N (\text{population size}) / S (\text{area})$$



POPULATION GROWTH

POPULATION GROWTH

- **The population is the total number of people in a given area, and population growth is the change in the measure of the living people in a nation.**
- **Population growth can be positive or negative, based on the number of deaths and births if a country has increased birth rates over death, the population increases, while more ends lead to a decrease.**
- **There are 7.7 billion people on this planet.**

POPULATION GROWTH CAN BE **UNDERSTOOD BY FOUR MAJOR POINTS:**

- 1. AN EARLY PERIODS OF HUNTERS AND GATHERERS***
- 2. EARLY, PRE-INDUSTRIAL AGRICULTURE***
- 3. THE AGE OF INDUSTRIAL REVOLUTION***
- 4. THE MODERN ERA***

1. AN EARLY PERIODS OF HUNTERS AND GATHERERS

This period ranges between the first evolution of humans on this planet to the beginning of agriculture. During this period, it is estimated that the total population was probably less than a million, population density was about 1 person per 130 – 260 sq km in the most habitable areas, and the average rate of growth was perhaps less than 0.00011% per year.



2. EARLY, PRE-INDUSTRIAL AGRICULTURE

This period began some time between 9,000 BC and 6,000 BC and lasted approximately until the 16th century AD. The first major increase in population came during this period. The total human population of the world was approximately 100 million by 1 AD that further increased to about 500 million by 1600 AD. The population density increased greatly to about 1 or 2 people per sq km or even more. The average rate of growth was probably about 0.03%, which was large enough to increase the human population from approximately 5 million in 10,000 BC to about 100 million in 1 AD.



3. THE AGE OF INDUSTRIAL REVOLUTION

The second and much more rapid increase in population started about 400 years ago with the industrial revolution associated with the advances in medicines and health care. Experts are of the opinion that Renaissance in Europe, (beginning about 1600 AD) marks the transition from agricultural to literate societies, when medical care and sanitation were factors responsible in reducing the death rate. The total human population of about 900 million in 1800 AD, almost doubled in the next



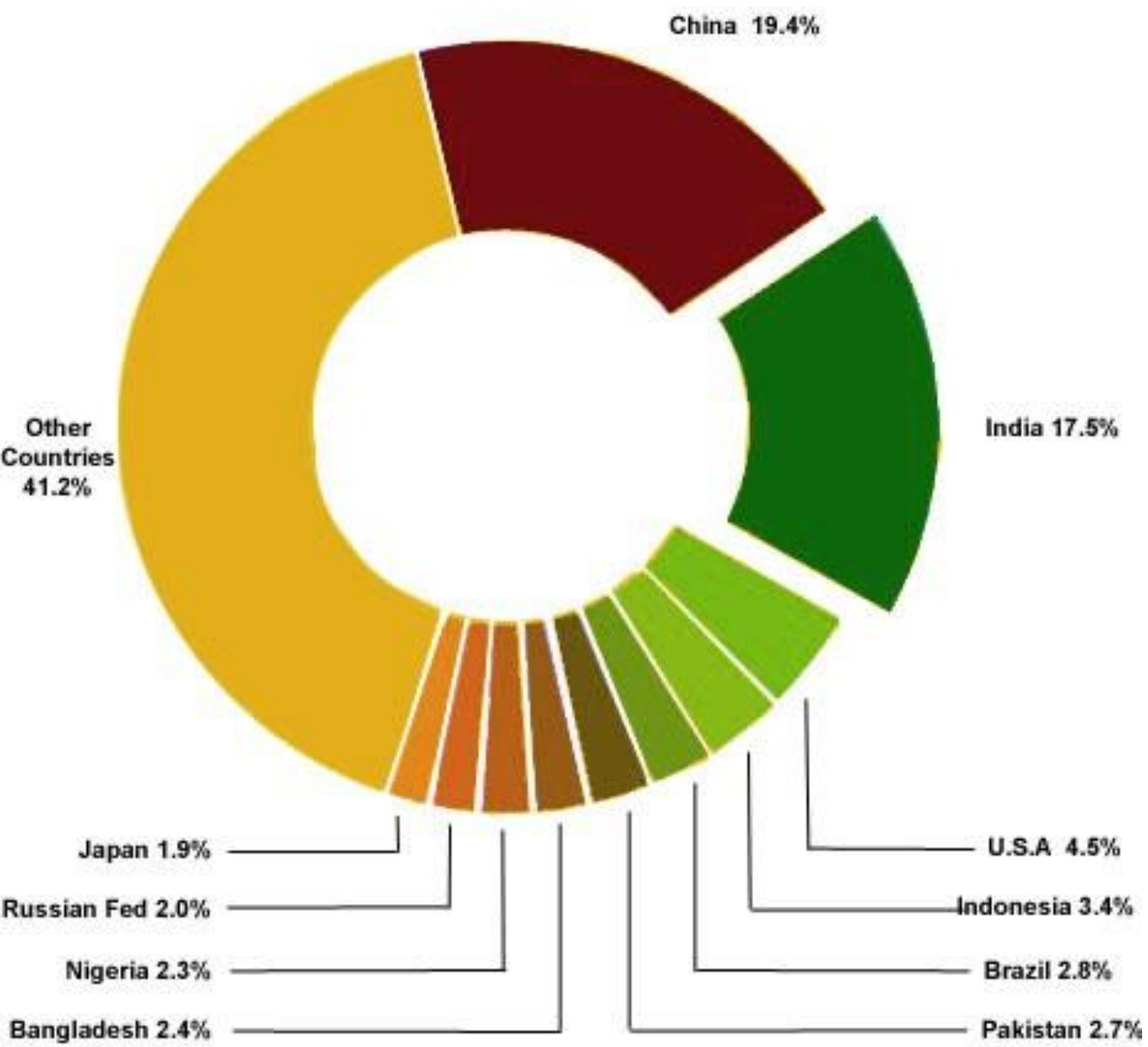
4. THE MODERN ERA

Though the rate of population growth has slowed down in developed countries, but population still continues to increase rapidly in many parts of the world, particularly developing countries. At mid 2002, the world population stood at 6.215 billions. The average rate of growth reached 2% in the middle of the 20th century and has declined to 1.3% in 2002 AD. According to projections, the global population will be approximately 8 billions by 2025 and 10 billions by 2050 AD.

According to World Bank projections, the world population may stabilize between 10.1 to 12.5 billions. Developed countries may increase from 1.2 billions today to 1.9 billions, but developing countries would increase from 5 billions to 9.6 billions. In these projections, the developing countries will contribute 95% of the increase.



DISTRIBUTION OF POPULATIONS



POPULATION EXPLOSION

Population explosion is a sudden spurt in the rate of population growth that occurs in the second stage of demographic transition due to a rapid fall in the mortality rate without a corresponding fall in the birth rate.



Population Exploitation

EFFECT OF HUMAN EXPLOSION

EFFECTS OF POPULATION EXPLOSION

- ✓ Over use of natural resources e.g. water, land, soil, natural forests.
- ✓ increase in food demand.
- ✓ increase in waste generation.
- ✓ increase demand of medical facilities.
- ✓ shortage of education facilities.
- ✓ Other effect like:-
 - ▶ unemployment
 - ▶ increase in crime rate

QUESTIONS FOR PRACTICES

1. Which of the sub-topic we had been completed?

- (a) Human Population
- (b) Human pollution
- (c) Both
- (d) None of these

Ans: (a)

2. How many types of Human Population?

- (a) One
- (b) Two
- (c) three
- (d) None of these

Ans: (b)

3. The formula for the human population

- (a) Population size x area
- (b) Population size + area
- (c) Population size - area
- (d) Population size/area

Ans: (d)

4. Population growth can be

- (a) Positive
- (b) Negative
- (c) Positive or Negative
- (d) None of these

Ans: (c)

5. Ecological density is the type of

- (a) Human population
- (b) Human density
- (c) Both
- (d) None of these

Ans: (b)